

Hathor - Weekly Report - Week 1

Summary

During the period of July 11 to July 17, the team focused on Milestone 0 (M0), establishing the core system architecture, data contracts, security guidelines, and delivery roadmap for the Hathor platform. The primary goal was to eliminate structural, security, and transport ambiguities before initiating active service implementation. Major achievements include approving the system architecture, defining microservice boundaries, finalizing HTTP and event contracts, establishing security and reliability standards, and formulating the overall Execution Plan.

Completed Tasks

System Architecture & Service Boundaries Approved

- **Status:** Completed
- **Microservice Boundaries:** Defined clear operational boundaries for five primary services (API Gateway, Auth Service, Catalog Service, Commerce Service, and Library Service). Established that each domain service maintains exclusive ownership of its own data store without cross-database reads or joins.
- **Local Container Topology:** Approved local-first Docker Compose topology comprising four physical PostgreSQL database instances, RabbitMQ message broker, Redis key-value cache, Memcached, API Gateway proxy, and Web application shell on private container networks.
- **Decision Records:** Finalized Architectural Decision Record ADR-001 establishing the local-first August release scope and architectural guardrails.

HTTP, Event, & Data Contracts Standardized

- **Status:** Completed
- **OpenAPI Contracts:** Standardized and approved public-api.openapi.yaml for external routes and internal-api.openapi.yaml for private inter-service communication.
- **AsyncAPI Event Contracts:** Approved domain-events.asyncapi.yaml specifying message formats for commerce outbox events, library entitlement inbox events, and payment status updates.
- **Canonical Models:** Designated gameId as the canonical game identifier across all domain services, database schemas, and event payloads. Formatted financial line items with exact decimal precision in EGP currency.
- **Validation Pipeline:** Selected OpenAPI and AsyncAPI schema linting tools to enforce strict contract compliance across all service builds in CI.

Security, Reliability, & Runbooks Established

- **Status:** Completed
- **Security Controls:** Approved Argon2id credential hashing, short-lived RS256 Access JWTs with public JWKS key distribution, rotating HttpOnly refresh token cookies, and Cloudflare Turnstile bot detection integration.
- **Inter-Service Authentication:** Established 5-minute scoped internal service JWTs for secure private service-to-service HTTP communications.
- **Reliability & Resilience:** Approved transactional outbox patterns for commerce, idempotent inbox processing for library entitlements, retry policies, and Dead Letter Queue (DLQ) replay mechanisms.
- **Operations & Runbooks:** Finalized operational runbooks for initial admin bootstrap, DLQ message replay, and database backup and restore procedures.

Execution Plan & Delivery Roadmap Formatted

- **Status:** Completed
- **Governance Framework:** Formatted the 7-milestone delivery roadmap (M0 Architecture & Contracts -> M1 Local Platform & Web Shell -> M2 Catalog & Cart -> M3 Payments & Entitlements -> M4 Downloads & Creator AI -> M5 Integration & Recovery -> M6 Release Hardening).
- **Task Definitions:** Established clear Definitions of Ready and Definitions of Done to govern issue progression across team boards.
- **Teammate Allocations:** Assigned primary ownership across 4 teammates covering Platform/Gateway (Teammate 1), Auth/Commerce (Teammate 2), Catalog/Library (Teammate 3), and Web Shell (Teammate 4).

Next Week Objectives

1. Initialize the monorepo workspace and multi-container Docker Compose infrastructure (Issue 1.1).
2. Embed contract validation tooling into the CI testing pipeline (Issue 1.2).
3. Implement Gateway edge routing, security middleware, and rate limiting (Issue 1.3).
4. Implement Auth service password hashing, RS256 JWT issuance, and session refresh cookies (Issue 1.4).